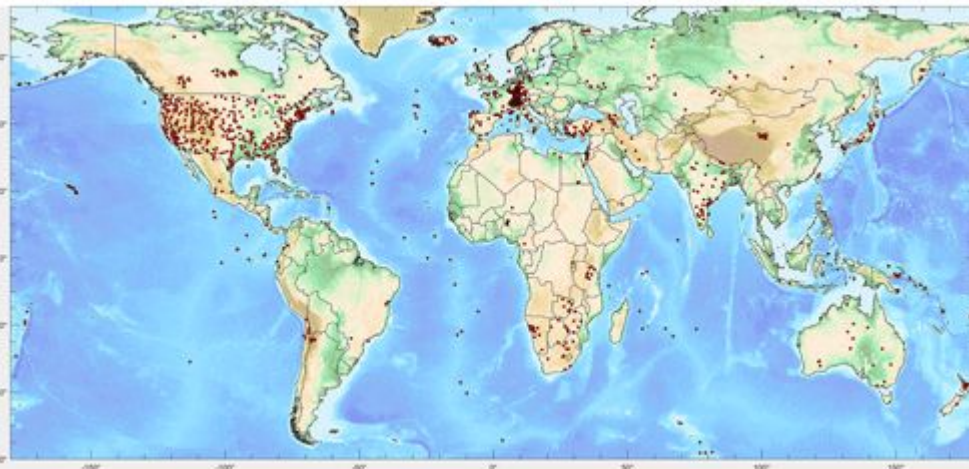




Critical Minerals and Net Zero

Rohan Nanda



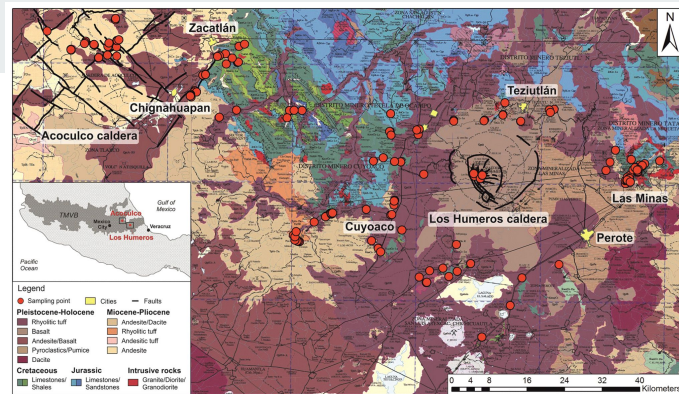
ESSA: Earth Systems Science Data

The PetroPhysical Property Database (P₃) – a global compilation of lab-measured rock properties

Kristian Bär, Thomas Reinsch, and Judith Bott
 Earth Syst. Sci. Data, 12, 2485–2515,
<https://doi.org/10.5194/essd-12-2485-2020>, 2020

Un na me d:	Longitude	Latitude	Temperature_K_min	Temperature_K_max	GrainDensity_kg_m3_min	GrainDensity_kg_m3_max	PWaveVelocity_m_s_min	PWaveVelocity_m_s_max	SWaveVelocity_m_s_min	SWaveVelocity_m_s_max	RockConductivity_S_m_min	RockConductivity_S_m_max	RockResistivity_M_ohm_min	RockResistivity_M_ohm_max
1	0													
2	17	-158.5966	21.48788	293		2890	3290	4810	6450	2780	3680			
3	18	-158.00059	21.43551	293		2850	3120	4140	7010	2330	3440			
4	19	-157.90736	21.35514	293				5350	7940	2970	3420			
5	20	-157.0271	20.89171	293				4450	6610					
6	21	-156.33146	20.79594	293		2680	3320	3400	6640	2130	3580			
7	22	-156.32122	20.7017	293		3139	3139					0.018181818	0.069444	14.4
8	23	-155.86669	19.69198	293				5570	8240	2910	3190			55
9	24	-155.51027	19.58599	293		2570	3280	2810	6650	1440	3500			
10	25	-155.28371	19.40622	293				4670	4930	2680	2700			
11	26	-153.432589	59.3622	298.15	298.15	2491	2556							
12	31	-123.504009	50.6315					2000	3090					
13	48	-122.189675	46.19978	298.15	1273.15	2642	2661							
14	68	-120.65895	43.9603	293								0.043	0.046	21.6

Rock Type (bulk)	weight % quartz for ultramafic =	weight % quartz for thermal metamorphism =	Particle density [g/cm ³]	Standard deviation particle density [g/cm ³]	Particle volume [cm ³]	Standard deviation particle volume [cm ³]	Bulk density [g/cm ³]	Standard deviation bulk density [g/cm ³]	Bulk volume [cm ³]	Standard deviation bulk volume [cm ³]	Petrophysical properties: density, porosity and permeability						Mean value (for conductivity [W/m])
											Porosity measured [%]	Standard deviation porosity [%]	Porosity interpolated [%]	Porosity calculated [%]	Note	Porosity permeability [mD]	
79 Chert	59.543	1.55	0.0002	23.535	0.0005	2.66	0.0048	23.535	0.0047	0.34	0.01	0.34	0.34	1.17 calculated using saturated weight, 50%	1.952		
80 Chert	70.749	1.55	0.0002	23.535	0.0005	2.70	0.0027	26.707	0.0022	0.36	0.01	0.36	0.36	0.77 calculated using saturated weight, 50%	3.556		
81 Chert	85.303	2.65	0.0009	31.624	0.0050	2.68	0.0071	31.651	0.0029	0.09	0.02	0.1	0.1				
82 Chert	240.038	2.65	0.0009	31.668	0.005	2.68	0.0058	34.822	0.0029	1.18	0.05				2.786		
83 Chert	31.660	2.65	0.0017	34.713	0.0115	2.65	0.0058	34.822	0.0029	1.18	0.05				2.786		
84 Chert	223.530	2.65	0.002	56.741	0.0035												
85 Chert	65.145	2.65	0.0009	32.579	0.0050	2.64	0.0024	32.231	0.0024	0.23	0.05	0.1	0.1		2.716		
86 Chert	243.352	2.65	0.0032	31.277	0.003												
87 Chert	67.000	2.65	0.0009	34.862	0.006	2.64	0.0024	34.876	0.0017	0.53	0.05	0.1	0.1		2.716		
88 Chert	65.317	2.65	0.0004	40.759	0.0097	2.58	0.0025	41.206	0.0470	1.07	0.01			1.10 calculated using saturated weight, 50%	2.476		
89 Chert	184.607	2.65	0.0002	41.499	0.0030	2.53	0.006	43.460	0.1027	0.09	0.01	0.1	0.1	0.84 calculated using saturated weight, 50%	1.952		
90 Chert	70.295	2.65	0.0009	26.7638	0.0033	2.62	0.0033	26.7447	0.0048	0.59	0.05	0.1	0.1	0.84 calculated using saturated weight, 50%	3.556		
91 Chert	2.65	0.0009	24.463	0.004			0.0043	24.276	0.0458	0.06	0.10	0.1	0.1				
92 Limestone	2.65	0.004	68.8722	0.323													
93 Limestone	2.65	0.004	67.5330	0.166													
94 Limestone	2.65	0.0039	66.5138	0.238													
95 Limestone	2.65	0.0044	60.9337	0.178													
96 Limestone	2.65	0.0019	57.584	0.195													
97 Limestone	2.65	0.001	16.054	0.023													
98 Limestone	2.65	0.0043	103.5272	0.950													
99 Limestone	2.65	0.004	63.5256	0.817													
100 Limestone	55.078	2.65	0.0007	21.042	0.0025	2.83	0.0043	20.3843	0.0350	0.30	0.02	0.1	0.1	2.43 calculated using saturated weight, 50%			
01 Limestone	2.65	0.0002	32.579	0.003	2.62	0.0095	26.026	0.0550	0.40	0.10	0.1	0.1	0.1				
02 Limestone	2.65	0.0004	32.579	0.003	1.94												
03 Gneiss/Gneissite	248.348	2.65	0.0002	34.446	0.003												
04 Gneiss/Gneissite	63.345	2.65	0.0003	35.8844	0.0072	2.47	0.0041	21.2867	0.0605	5.36	0.05						
05 Gneiss/Gneissite	282.353	2.65	0.002	50.5891	0.2070												
06 Gneiss/Gneissite	253.362	2.65	0.0002	36.7438	0.003												
07 Limestone	277.640	2.65	0.0006	34.2905	0.003												
08 Limestone	263.740	2.65	0.0006	35.5842	0.003												
09 Limestone	273.628	2.65	0.0003	33.7458	0.003												
10 Limestone	244.176	2.65	0.0006	32.3807	0.003												
11 Limestone	278.794	2.65	0.0008	34.6459	0.003												
12 Limestone	272.847	2.65	0.0004	30.3010	0.003												
13 Limestone	280.448	2.65	0.0017	105.7239	0.9670												
14 Limestone	344.353	2.65	0.0009	74.6594	0.0339												



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Rock Type	Mean elec	Mean elec	Mean elec	Mean electric resistivity sat	[Ω m]	max			Mean P-w	Mean P-w	Mean S-w	Mean S-w	Mean P-w	Mean P-w	Mean S-w	Mean S-w	Bulk dens	Bulk density	[g/cm ³]	max
2	Altered andesitic/dacitic dyke									4046	6160	2567	4194	5692	7957	3658	4664	2.6327	2.6597		
3	Altered andesitic/dacitic lava									2191	5654	1311	3347	2996	7705	1812	4451	2.0093497	2.6783		
4	Andesite			619.00773	737.87672	12007972				1511	6493	879	3731	2547	8170	1507	4671	2.0111332	2.786		
5	Andesite									3321	4532	2030	2807	4021	6330	2270	4017	2.5625	2.628		
6	Andesite Breccia									4822	4822	3048	3048					2.2824	2.2824		
7	Andesitic dyke									3121	5073	1918	2856	4725	6351	2734	3836	2.5513	2.6108		
8	Andesitic dyke																	2.4422	2.4422		
9	Andesitic ignimbrite									2733	2733			2763	2763			0.9615	0.9615		
10	Argillite/Argillitic limestone			206.00173	275.6110606202973					1759	6021	620	3191	2202	9389	1242	4205	2.6074	2.674		
11	Ash fall deposits									1719	2263	1083	1438	1925	2525	1178	1733	1.1164	1.4854		
12	Basalt			215.83814	215.8381492034545					2580	5458	1539	3430	3302	6738	2055	4041	1.9028	2.8122		
13	Basalt									3183	5659	1985	3160	4028	7225	2411	4369	2.1476	2.5365		
14	Basalt/Andesite									3456	4935	1819	2627	3289	5082	1764	2849	2.2763	2.6828		
15	Basaltic andesite									1703	6148	1009	3617	3418	7357	1853	4411	2.0237500	2.715077821822201		
16	Basaltic dyke									2427	6361	1467	3641	4017	8008	2334	4696	2.4361	2.9874		
17	Basaltic trachyandesite									2844	5779	1779	4061	3432	8253	2020	4750	2.4007	2.8185		
18	Chert									3678	7230	2167	4938	5945	9727	3523	5728	2.5564	2.697		
19	Dacite																	2.165	2.351		

Petrophysical and mechanical rock property database of the Los Humeros and Acoculco geothermal fields (Mexico)

Leandra M. Weydt, Ángel Andrés Ramírez-Guzmán, Antonio Pola, Baptiste Lepillier, Juliane Kummerow, Giuseppe Mandrone, Cesare Comina, Paromita Deb, Gianluca Norini, Eduardo Gonzalez-Partida, Denis Ramón Avellán, José Luis Macías, Kristian Bär, and Ingo Sass

Earth Syst. Sci. Data, 13, 571–598, <https://doi.org/10.5194/essd-13-571-2021>, 2021

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Formation	Density (g/cm³)	Density (g/cm³)	P-Wave Velocity (m/s)	P-Wave Velocity (m/s)	S-Wave Velocity (m/s)	S-Wave Velocity (m/s)	Velocity (m/s) Max		Electrical Resistivity (Ω·m)	Electrical Resistivity (Ω·m)		
2	CO ₂ Storage Sandstone	2	2.8	2000	4500	1000	2500			10	100		
3	Cobalt-Rich Oxides	4.5	5	5000	7000	3000	4000			0.1	10		
4	Copper Porphyry Deposits	2.6	2.8	4000	6000	2000	3500			10	1000		
5	Fractured Limestone	2.3	2.7	3000	6000	1500	3500			10	1000		
6	Graphite Schists	2.7	3	4000	6000	2500	3500			10	100		
7	High-Fracture Sandstones	2	2.5	2000	4000	1000	2500			10	100		
8	High-Porosity Aquifer	1.8	2.4	1500	3500	800	2000			1	100		
9	Lithium-Bearing Brines	1.2	1.3	1500	1500	1500	2500			0.2	0.5		
10	Manganese Nodules	2.3	2.5	2000	3000	1000	1500			1	10		
11	Nickel-Bearing Sulfides	4.5	5.5	6000	7000	3500	4000			0.1	5		
12	Platinum Group Element Deposits	4	5.5	5500	6500	3000	4000			0.1	10		
13	Rare Earth Element Deposits	2.5	3.5	4000	6000	2000	3500			10	500		
14	Water-Saturated Sands	1.8	2.2	1500	2500	800	1500			1	50		
15													
16													

USGS Publications

- Thermal properties, wave velocities, resistivity data./ Reports: *Thermal Properties of Rocks, Predicting S-Wave Velocities.*

SpringerLink & Elsevier Journals

- Studies on sandstone, aquifers, fractured limestone./ Example: "Multiscale Modeling of Compaction Bands."

Acta Geophysica

- Thermal and mechanical behaviors under varied conditions. / Example: "Variation of Wave Velocity and Porosity in Sandstone."

University Resources

- Electrical resistivity insights from academic studies. / Example: *Electrical Resistivity of Different Soil and Rock Types.*

Measurement conditions Problem



Critical Mineral Formations:

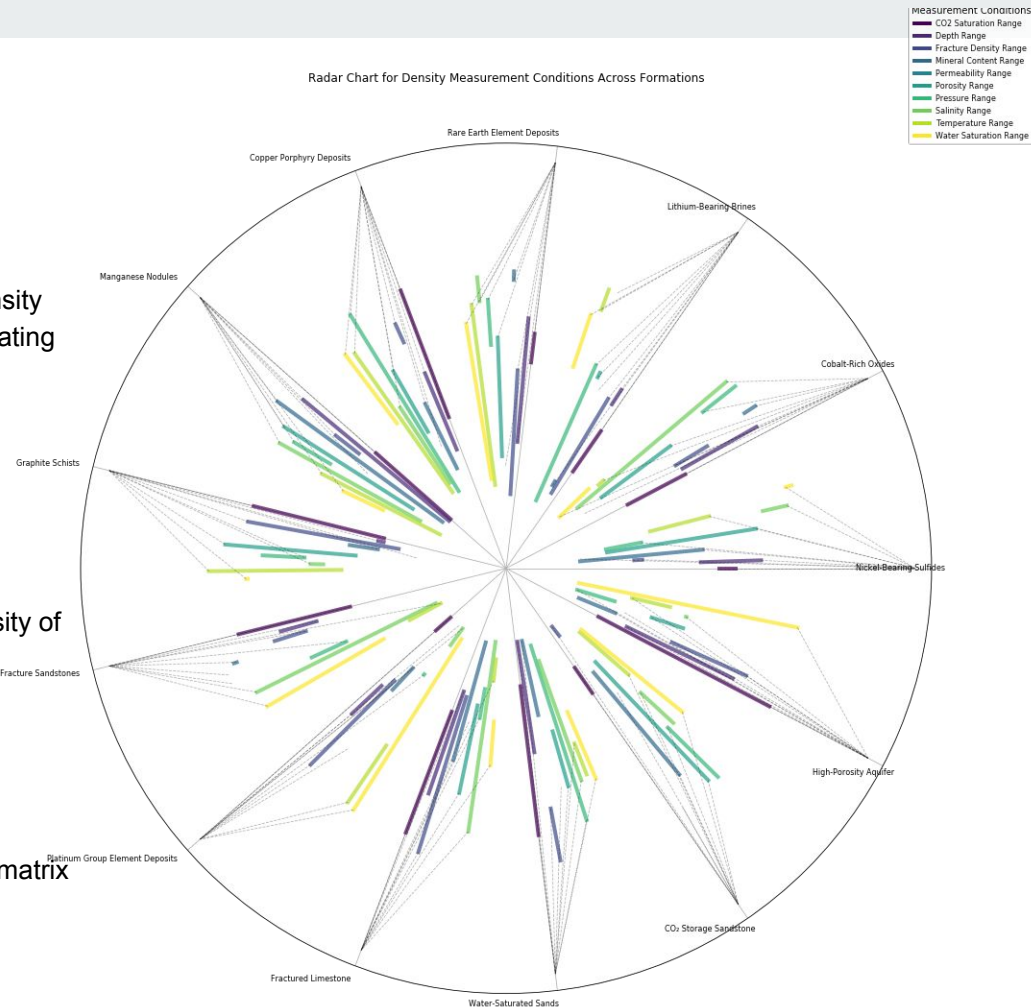
- **High Mineral Concentration** in cobalt-rich oxides increases density but overshadows the effects of pressure or temperature, complicating comparisons.
- **High Temperature** in rare earth element deposits reduces fluid density but minimally affects compacted solid density, creating inconsistencies.

CO₂ Reservoirs:

- **High CO₂ Saturation** reduces bulk density due to the lower density of CO₂ compared to water, masking the effects of pressure on rock compaction.

Water Reservoirs:

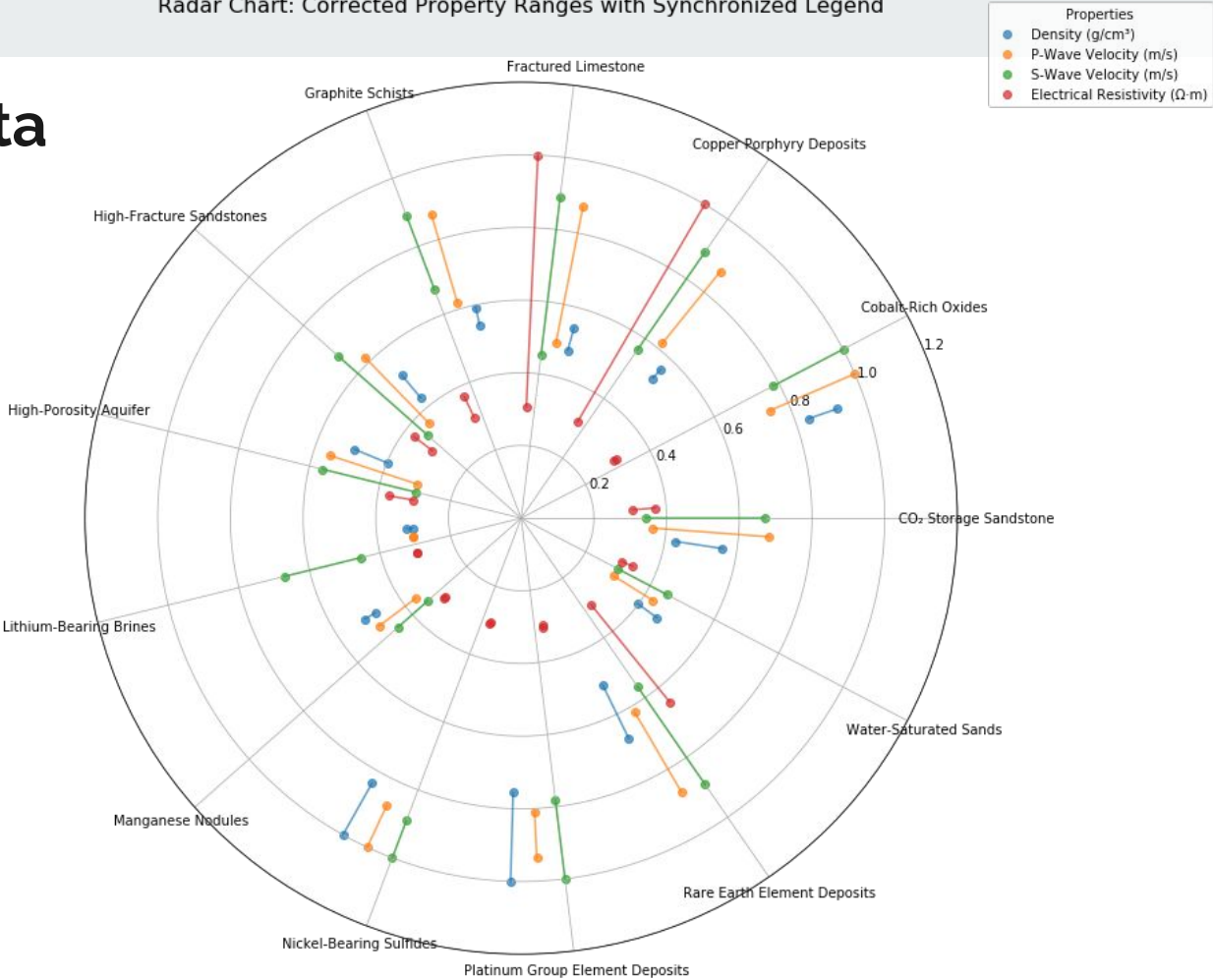
- **High Water Saturation** significantly increases density in porous aquifers, making it hard to separate fluid contributions from rock matrix properties.



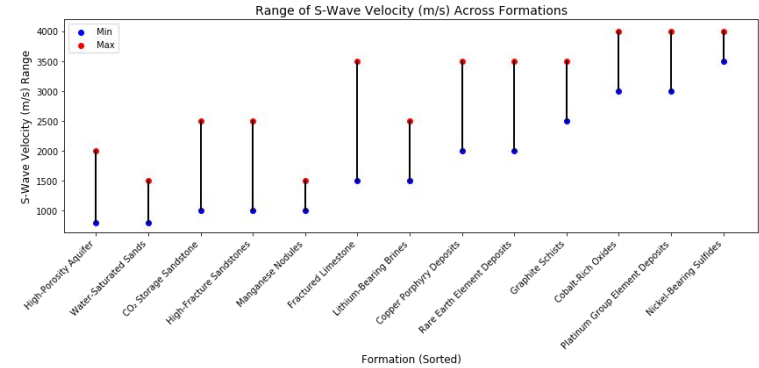
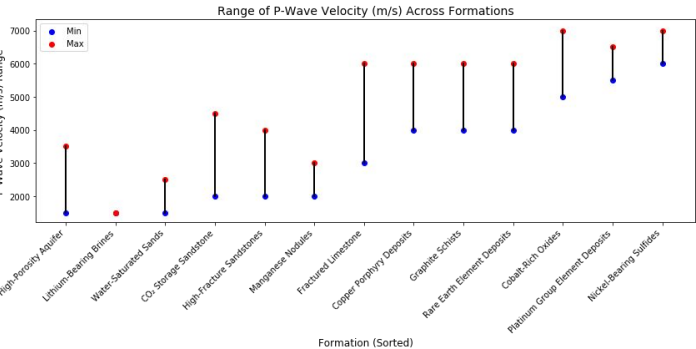
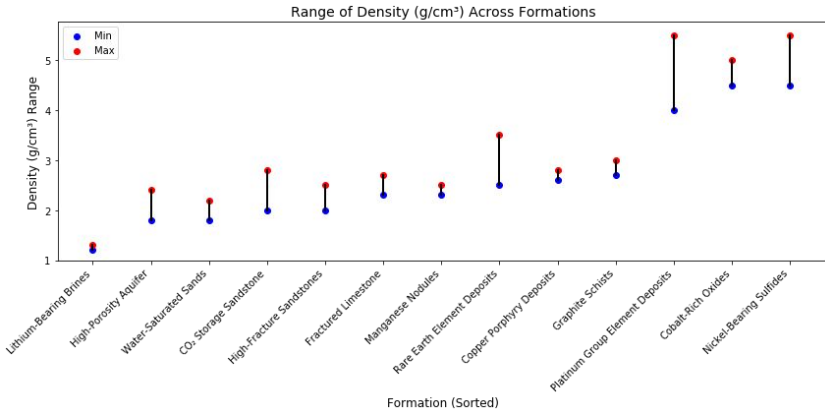
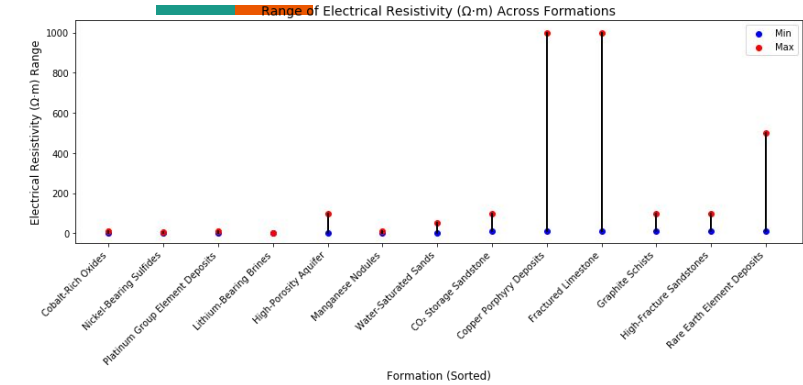
Formations Data



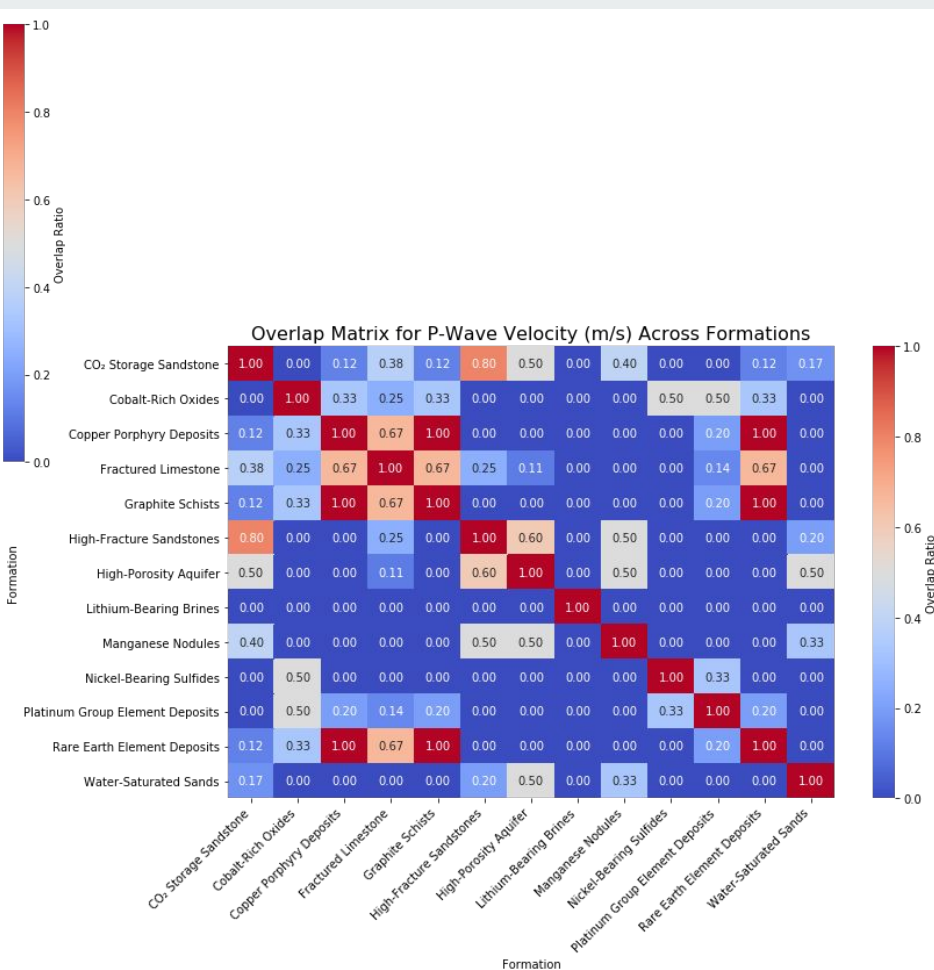
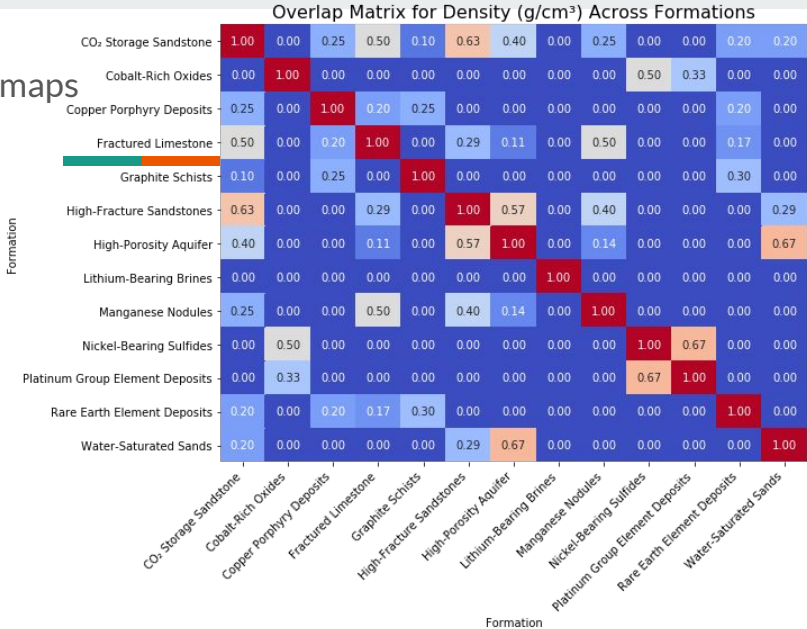
Radar Chart: Corrected Property Ranges with Synchronized Legend

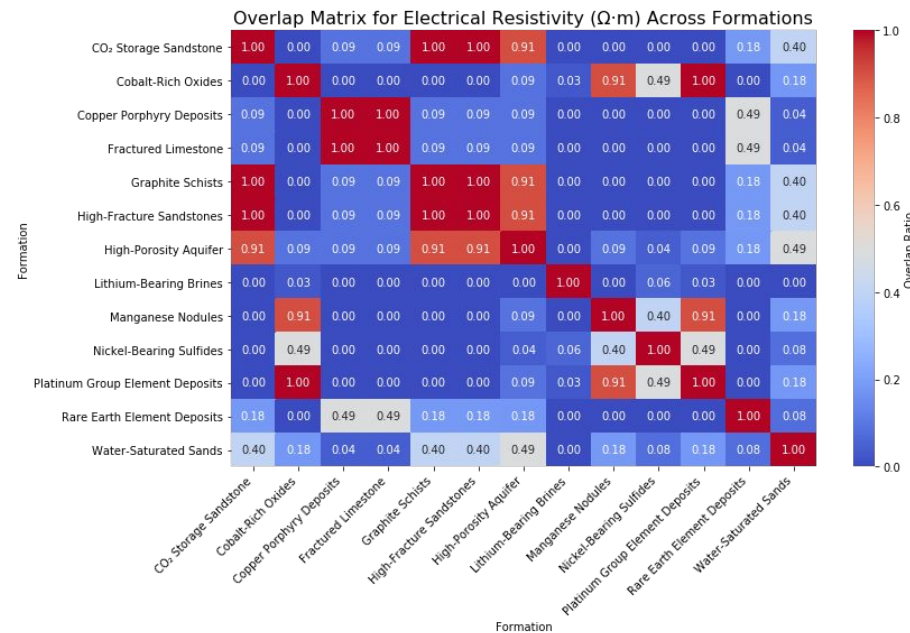
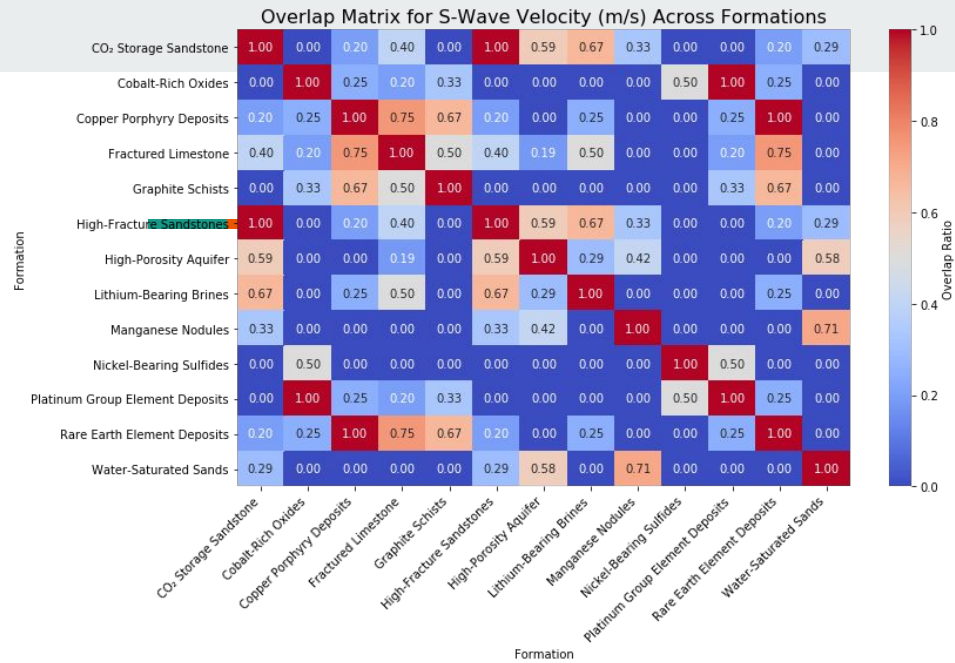


Trends



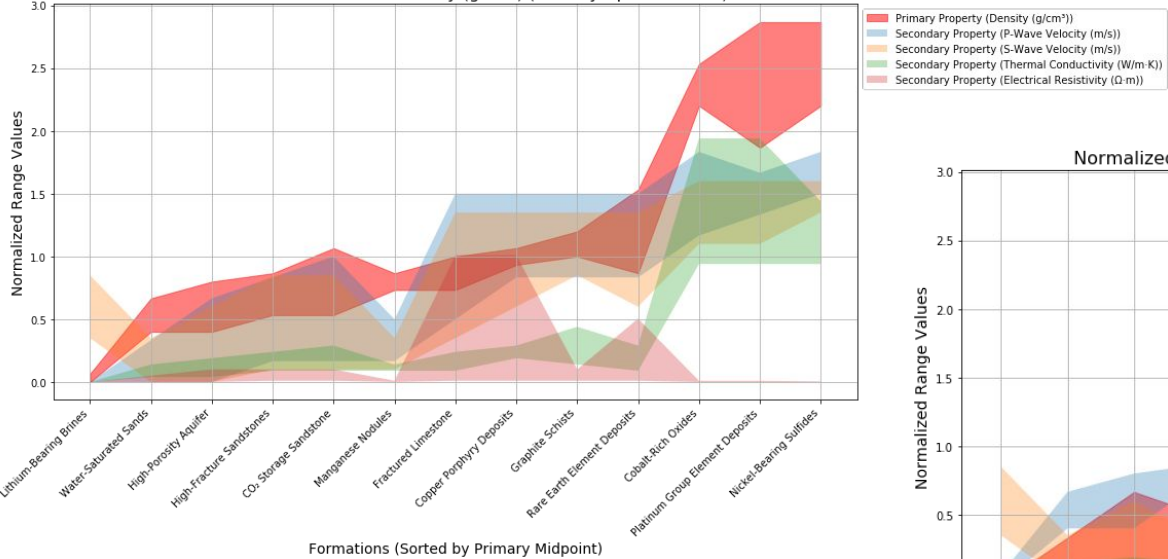
Heatmaps



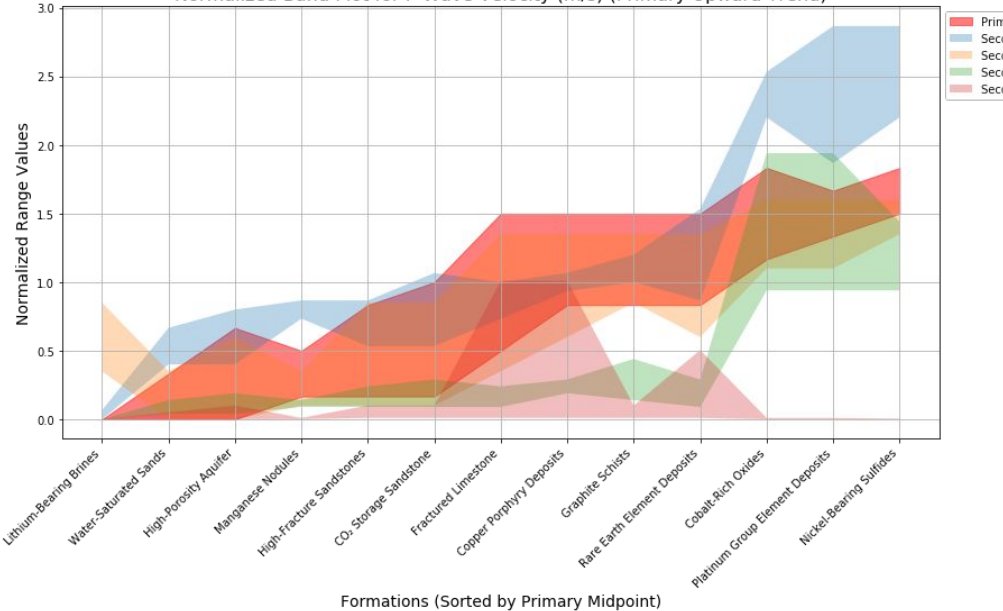


Formation trend comparison

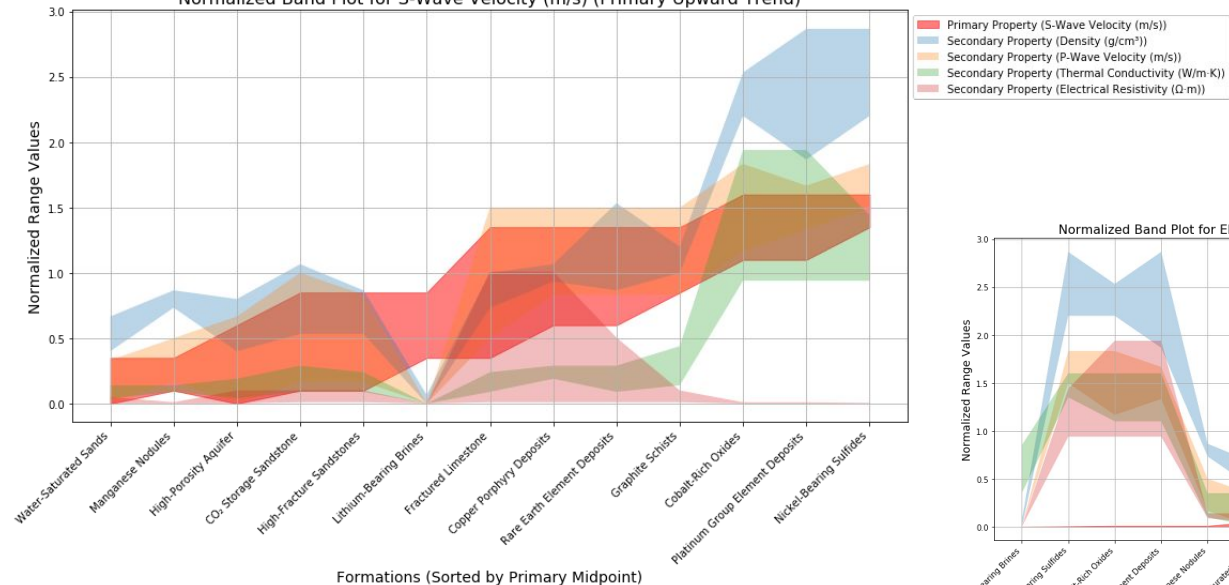
Normalized Band Plot for Density (g/cm³) (Primary Upward Trend)



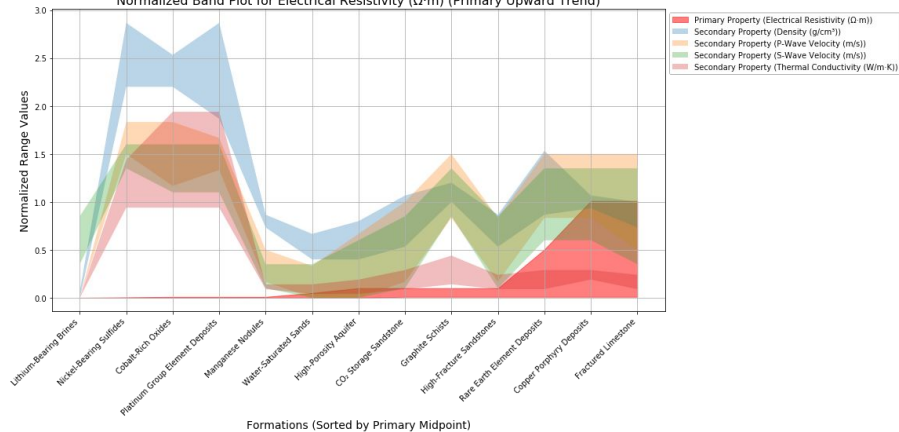
Normalized Band Plot for P-Wave Velocity (m/s) (Primary Upward Trend)



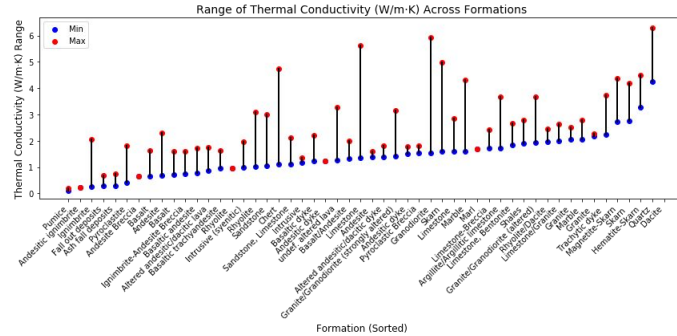
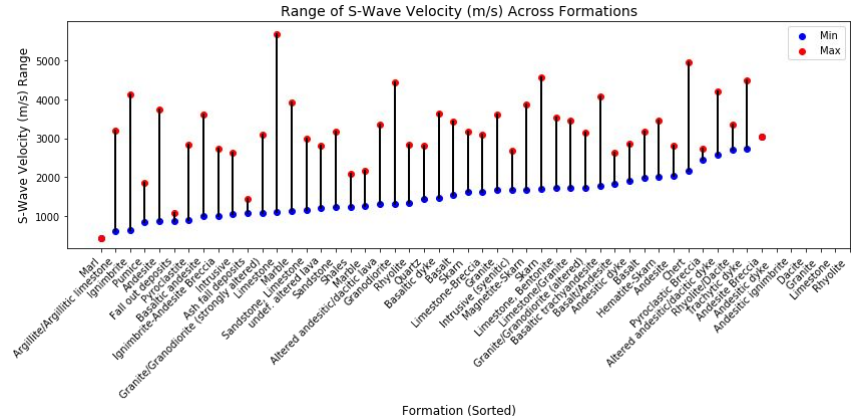
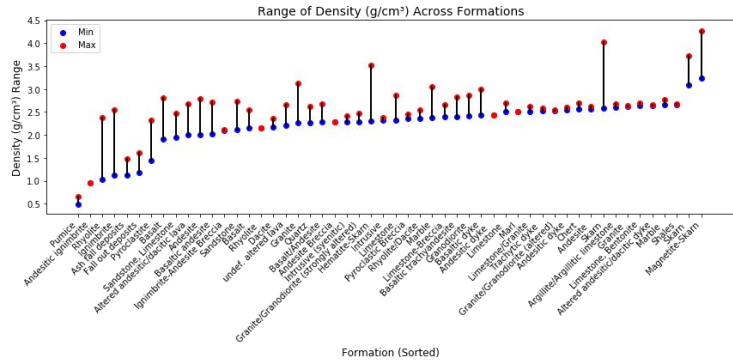
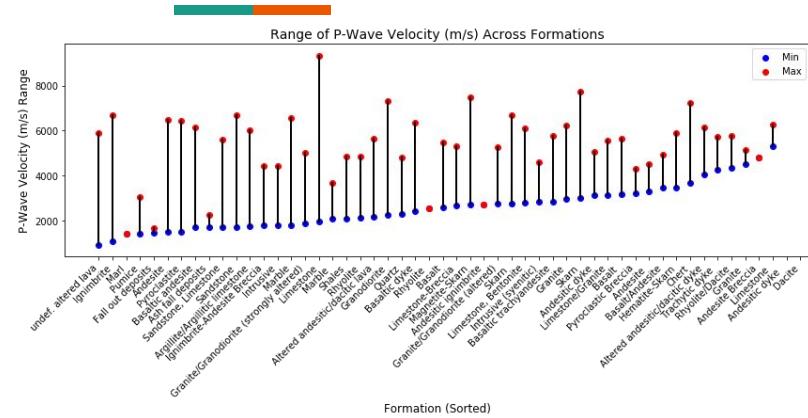
Normalized Band Plot for S-Wave Velocity (m/s) (Primary Upward Trend)



Normalized Band Plot for Electrical Resistivity (Ω·m) (Primary Upward Trend)

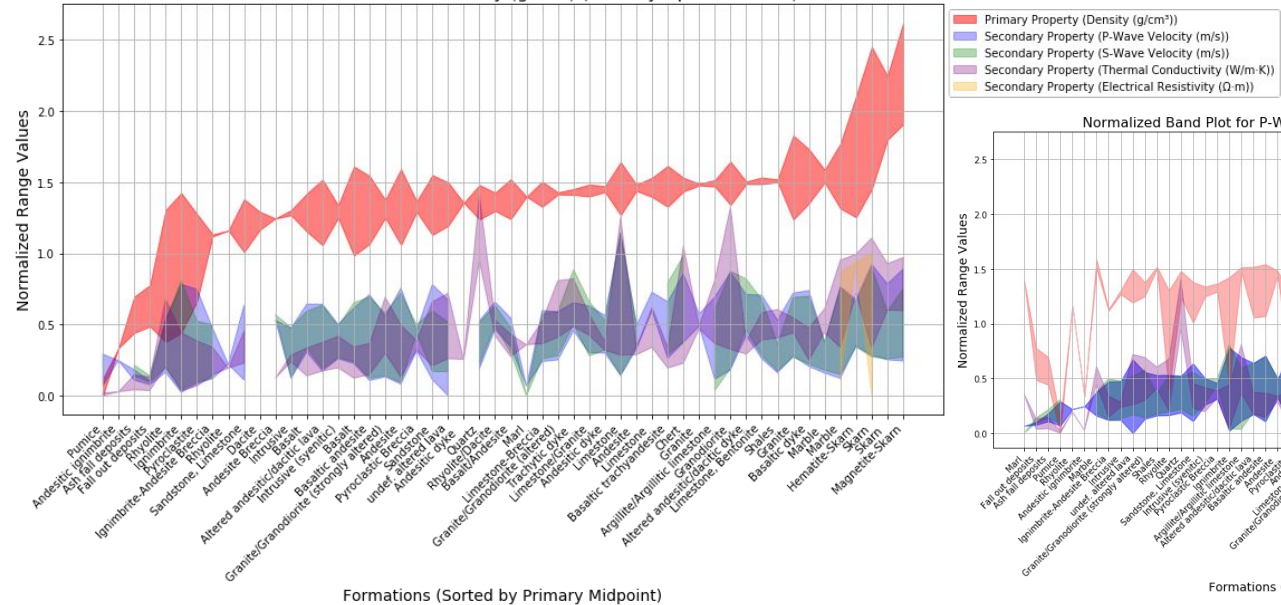


Mexico Data

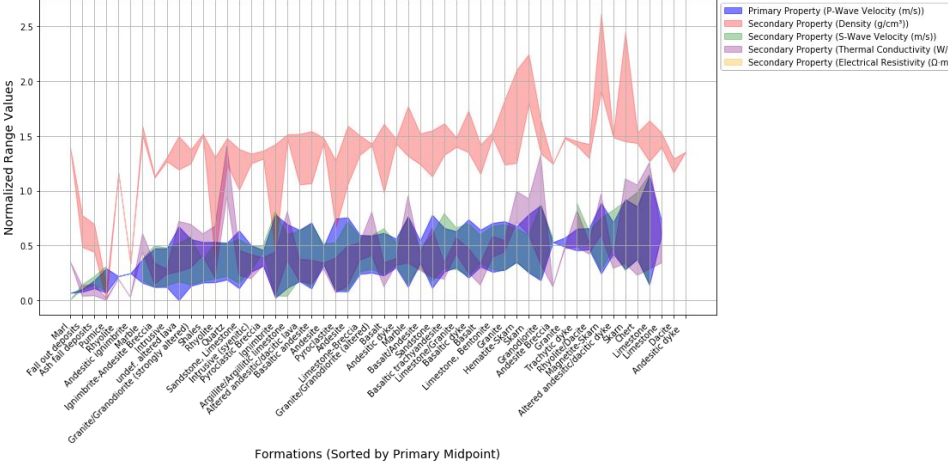


Mexico trend comparison

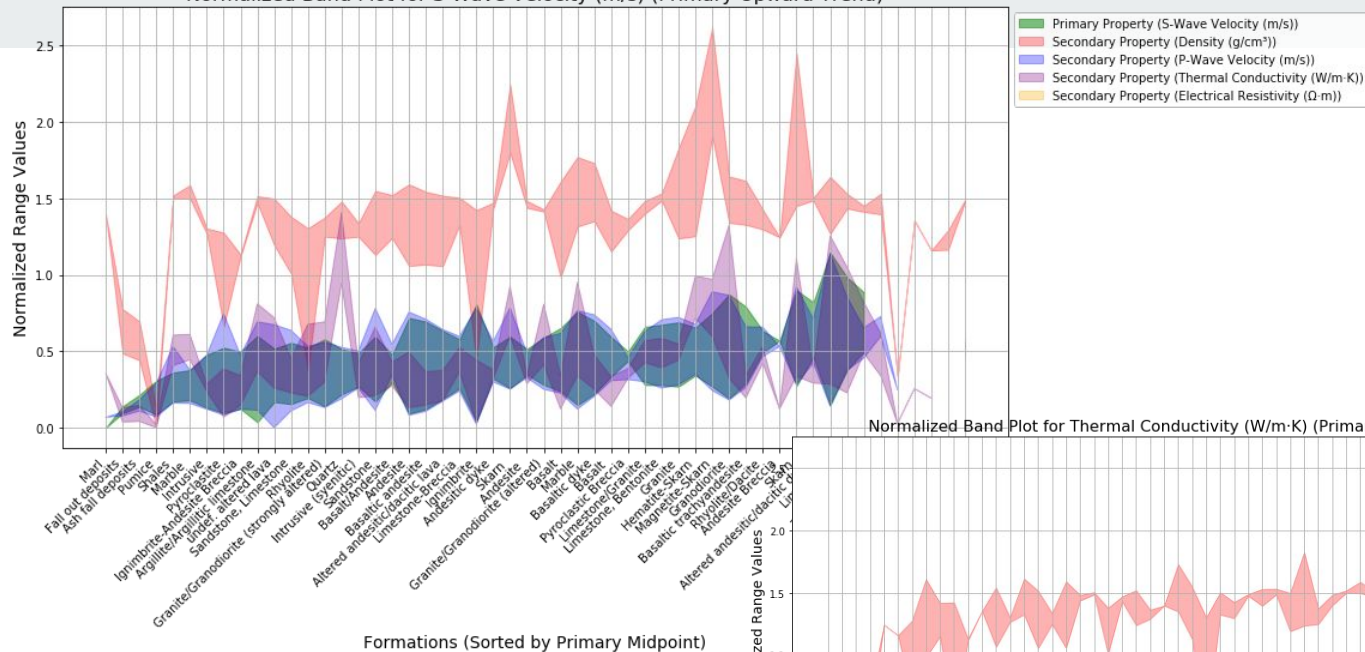
Normalized Band Plot for Density (g/cm³) (Primary Upward Trend)



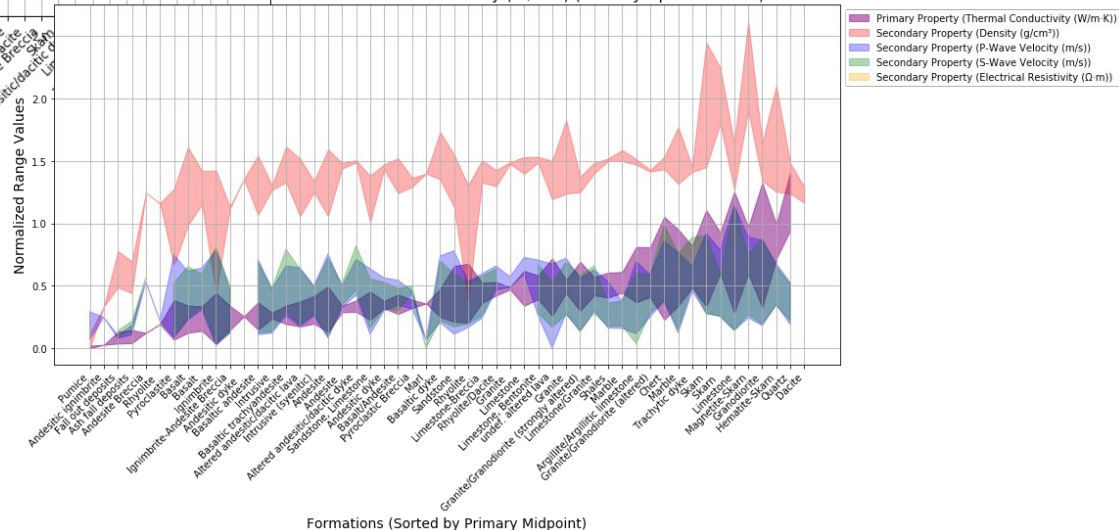
Normalized Band Plot for P-Wave Velocity (m/s) (Primary Upward Trend)



Normalized Band Plot for S-Wave Velocity (m/s) (Primary Upward Trend)



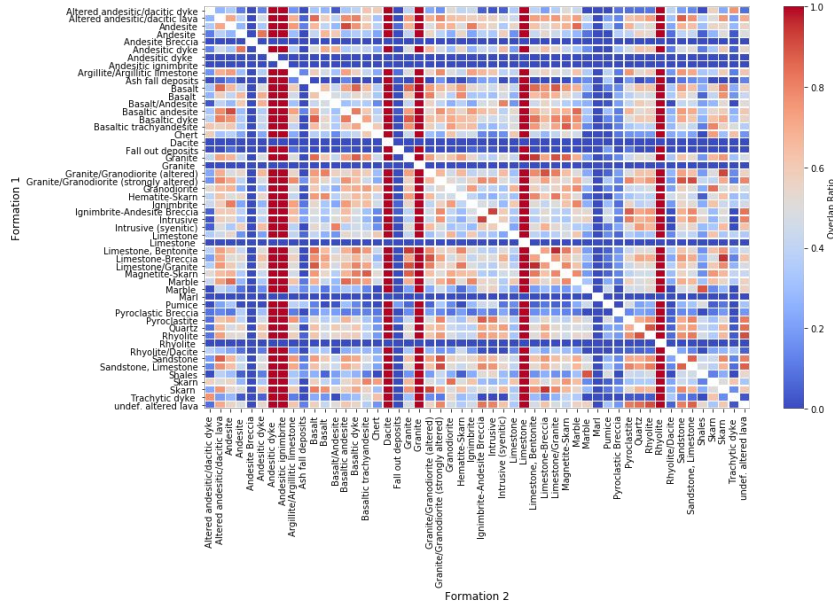
Normalized Band Plot for Thermal Conductivity (W/m-K) (Primary Upward Trend)



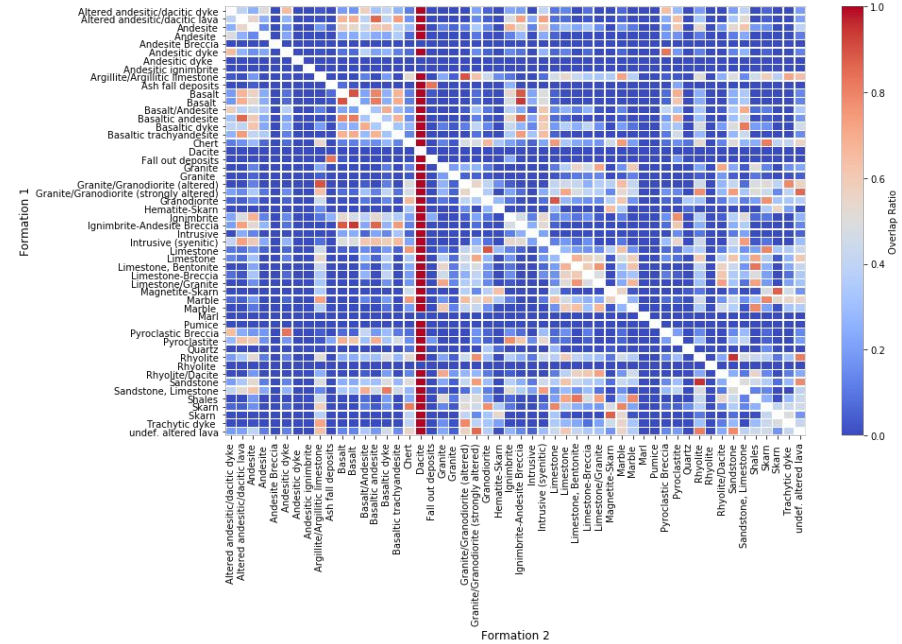
Mexico heatmaps



Heatmap of Overlap Ratios Between Formations for S-Wave Velocity (m/s)

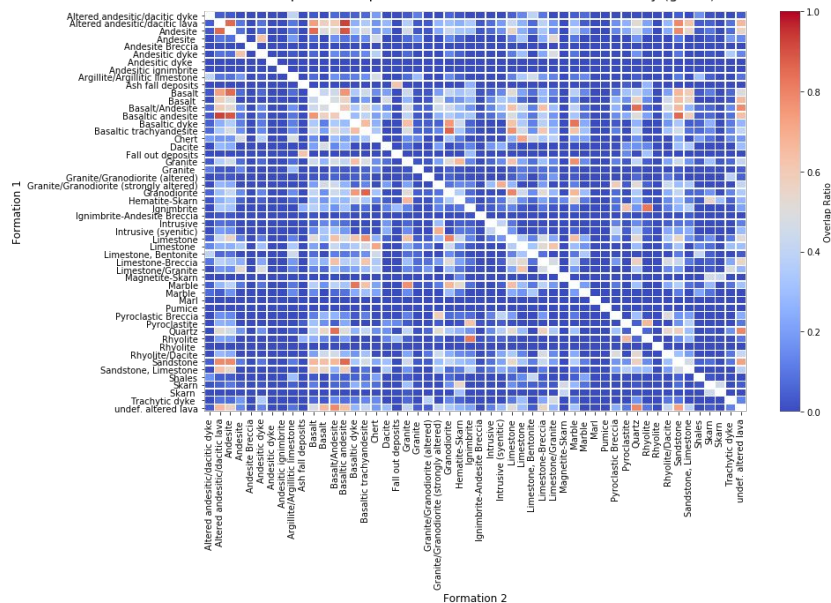


Heatmap of Overlap Ratios Between Formations for Thermal Conductivity (W/m-K)

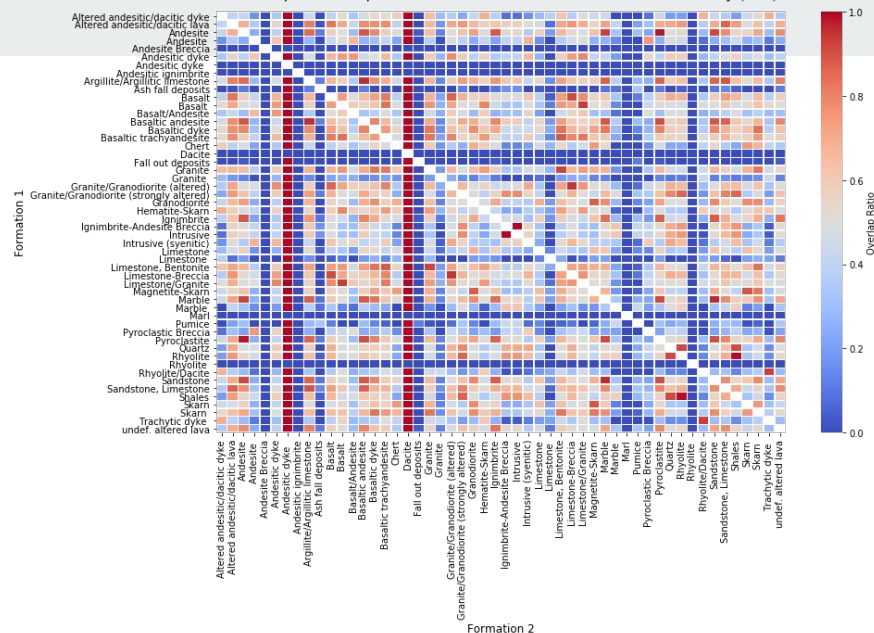




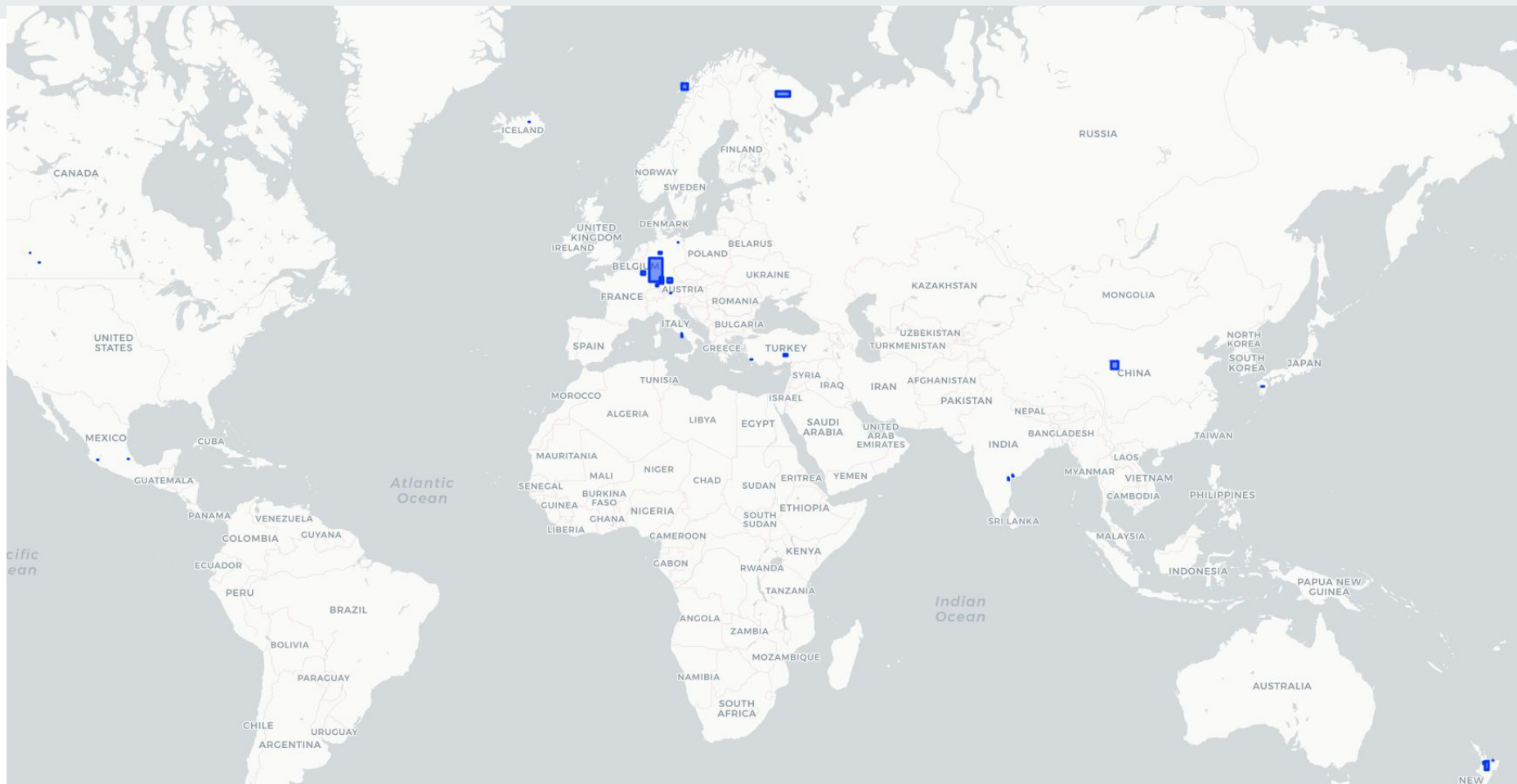
Heatmap of Overlap Ratios Between Formations for Density (g/cm^3)



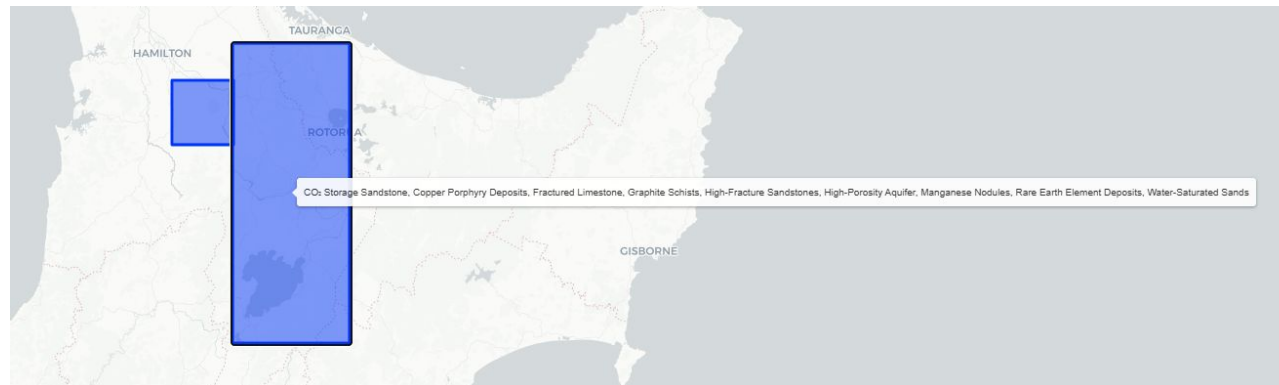
Heatmap of Overlap Ratios Between Formations for P-Wave Velocity (m/s)



Property Correlation.

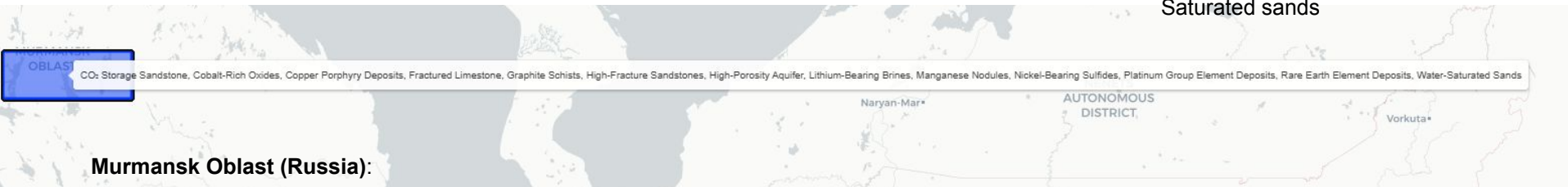


Property Correlation.



New Zealand:

- **Longitude Range:** 175.67819 to 176.33476
- **Latitude Range:** -39.06763 to -37.74275
- **Matches:** CO₂ Storage Sandstone, Copper Porphyry Deposits, Fractured Limestone, Graphite Schists, High Fracture Sandstones, High Porosity Aquifer, Manganese Nodules, Rare Earth Element Deposits, Water Saturated sands



Murmansk Oblast (Russia):

- **Longitude Range:** 32.92223 to 35.71895
- **Latitude Range:** 67.66991 to 68.05386
- **Matches:** CO₂ Storage Sandstone, Cobalt-Rich Oxides, Copper Porphyry Deposits, Fractured Limestone, Graphite schists, High fracture Sandstones, High-Porosity aquifer, Lithium bearing brines, Manganese Nodules, Nickel Bearing Sulfides, Platinum Group Element Deposits, Rare Earth Element deposits, Water saturated sands.

New Zealand Formations and Relevant Exploration

CO₂ Storage Sandstone

- **Model:** Bulk modulus (K_{eff}) for porous rocks: **Relevance:** Models porosity and fluid substitution for Taranaki sandstones.

$$K_{\text{eff}} = K_m + \phi \left(\frac{K_m - K_f}{1 + (1 - \phi) \frac{K_m - K_f}{K_f + 4/3 G_m}} \right)$$

Copper Porphyry Deposits

- **Method:** Induced polarization (IP) chargeability: **Relevance:** Identify Taupō arc deposits via IP anomalies.

$$M = \frac{(R_0 - R_{\infty})}{R_0}$$

Fractured Limestone

- **Model:** Shear modulus with microcrack density **Relevance:** Evaluates fractured karst aquifers in Waitomo.

$$G_{\text{eff}} = G_m \left(1 - \frac{4}{3} \pi \rho_c a^3 \right)$$

Graphite Schists

- **Technique:** Electromagnetic (EM) conductivity **Relevance:** Maps conductive Otago Schists.

$$\nabla \times \mathbf{H} = \sigma \mathbf{E}, \quad \nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

Rare Earth Element Deposits

- **Model:** Ion adsorption partition coefficient **Relevance:** Explores regolith-hosted REEs in Fiordland.

$$K_d = \frac{[REE]_{\text{solid}}}{[REE]_{\text{solution}}}$$